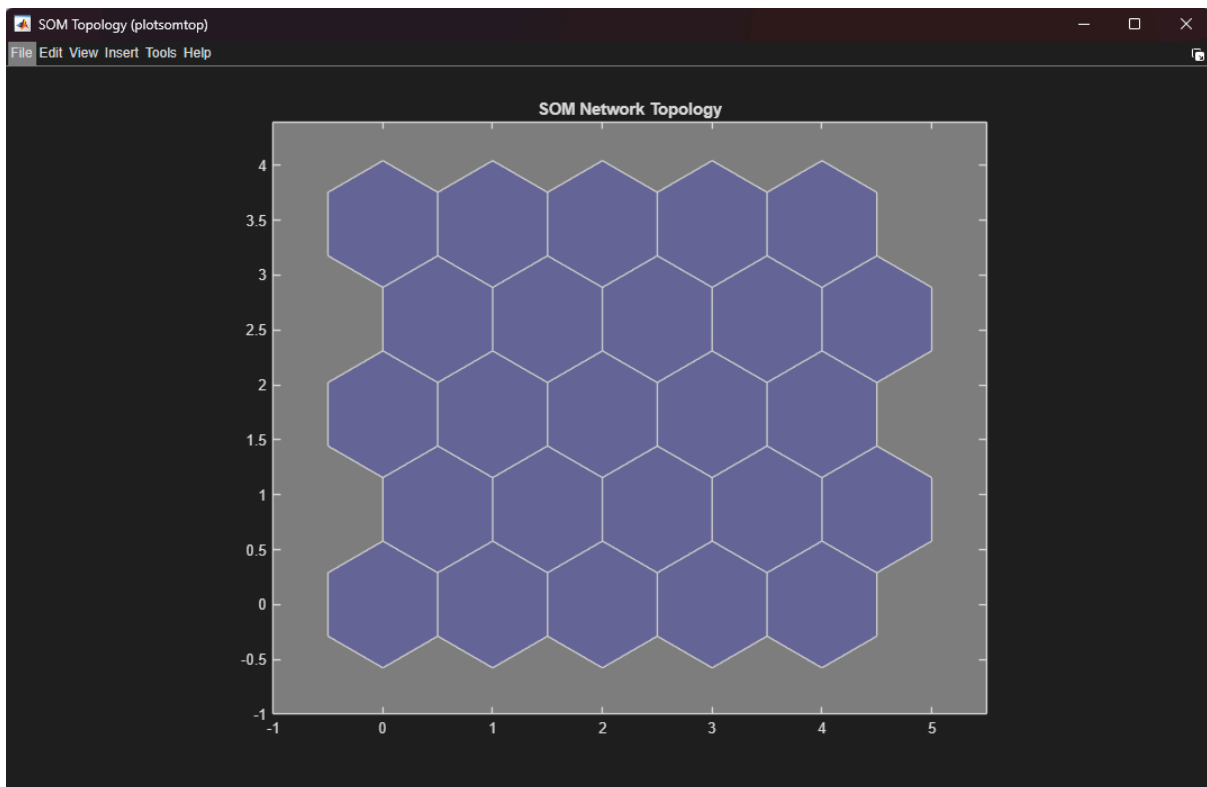
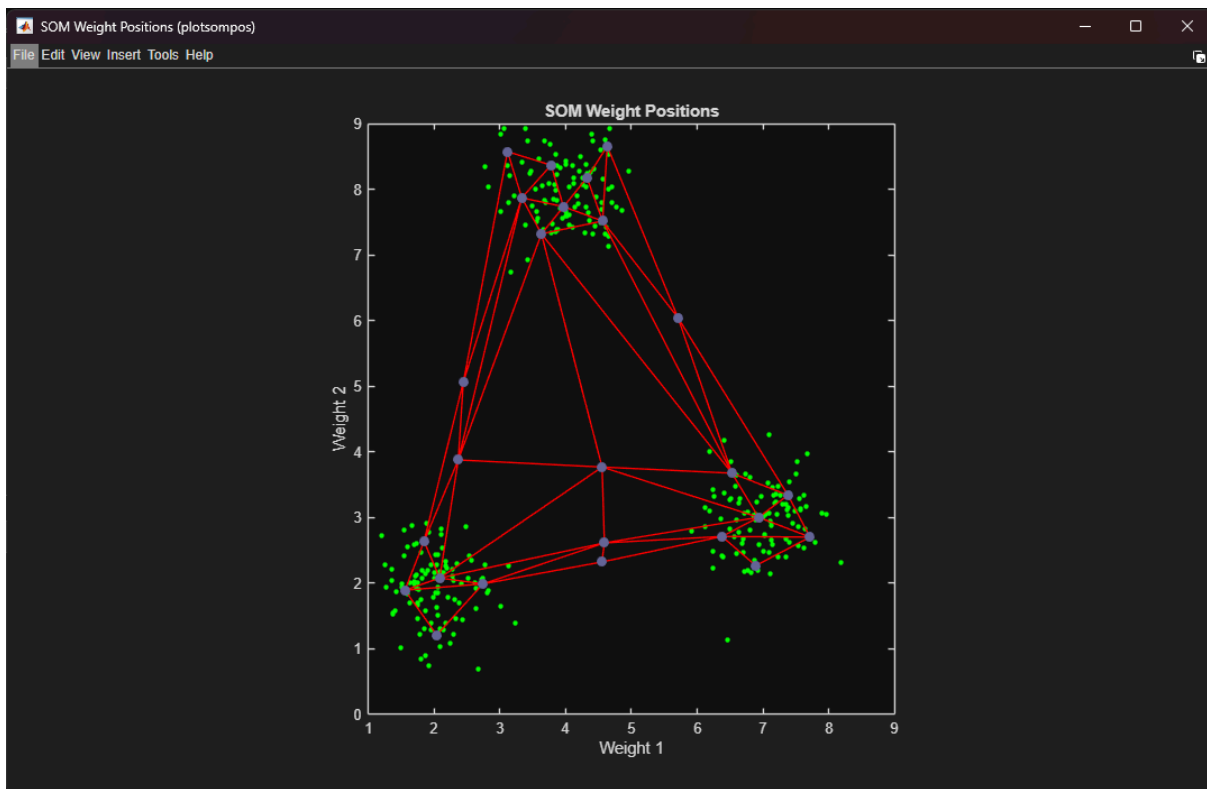
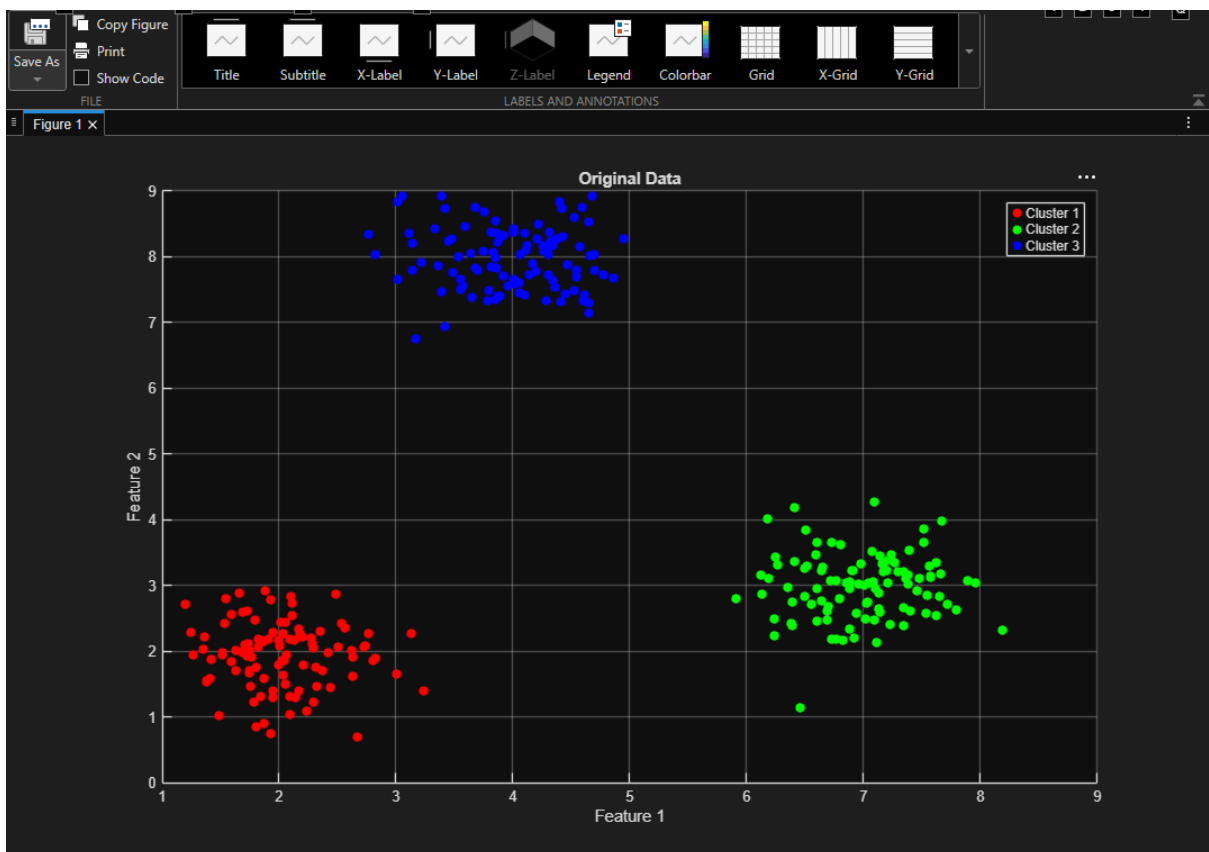
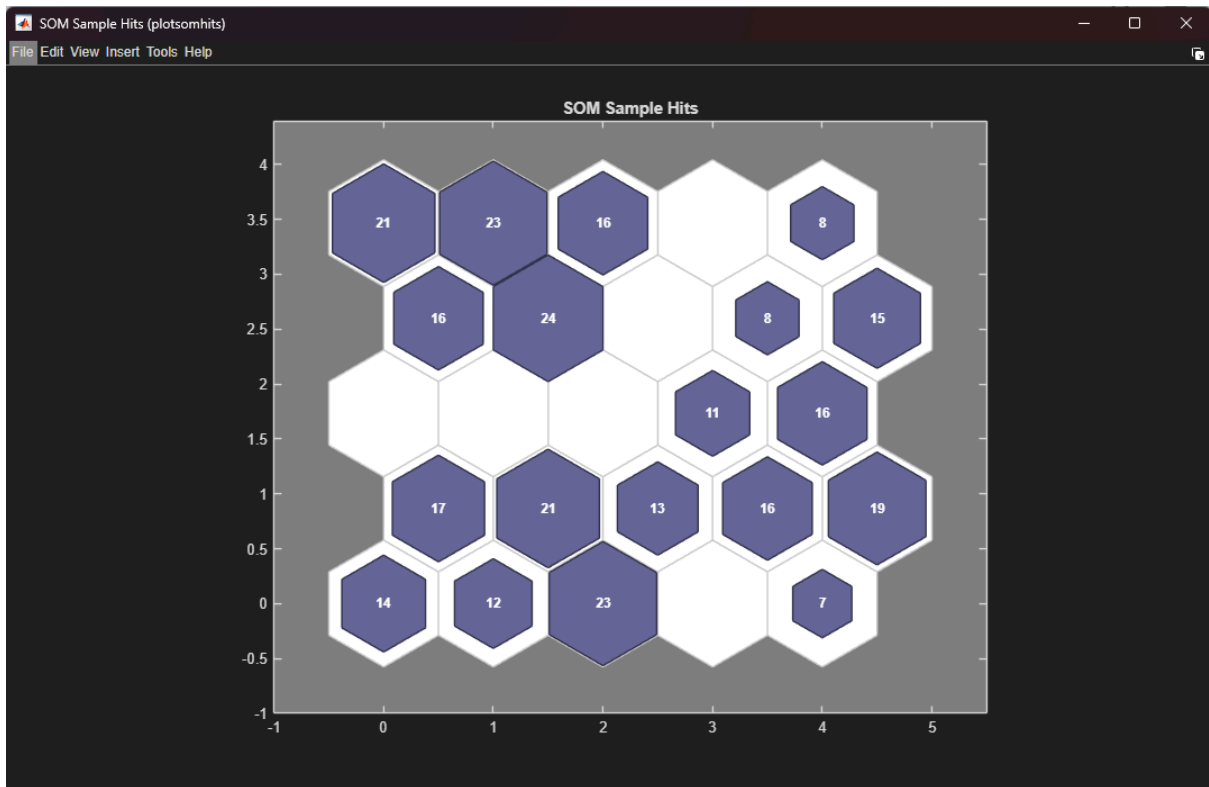
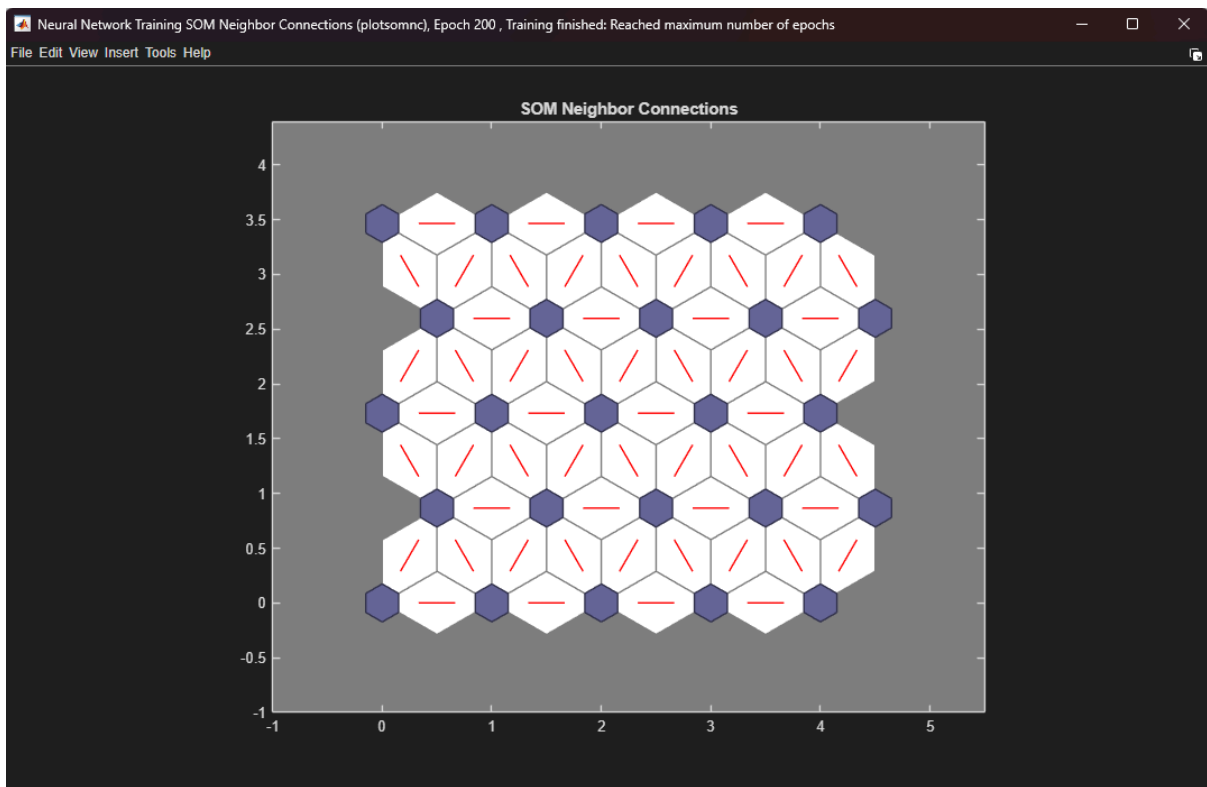


#	SOMClustering.m x	+
C:\Users\Afroz\Desktop\NN\EXP8\Code\SOMClustering.m		
1	clc;	
2	clear;	
3	close all;	
4		
5	%% Step 1: Generate clustered data	
6	rng(10); % For reproducibility	
7		
8	% Cluster 1	
9	C1 = randn(100,2) * 0.5 + [2 2];	
10		
11	% Cluster 2	
12	C2 = randn(100,2) * 0.5 + [7 3];	
13		
14	% Cluster 3	
15	C3 = randn(100,2) * 0.5 + [4 8];	
16		
17	% Combine all data	
18	data = [C1; C2; C3];	
19		
20	% SOM requires observations as columns	
21	X = data';	
22		
23	%% Step 2: Plot original data	
24	figure;	
25		
26	scatter(C1(:,1), C1(:,2), 40, 'r', 'filled');	
27	hold on;	
28		
29	scatter(C2(:,1), C2(:,2), 40, 'g', 'filled');	
30	scatter(C3(:,1), C3(:,2), 40, 'b', 'filled');	
31		
32	xlabel('Feature 1');	
33	ylabel('Feature 2');	
34	title('Original Data');	
35		
36	legend('Cluster 1','Cluster 2','Cluster 3');	
37	grid on;	
38		
39	%% Step 3: Create Self-Organizing Map	
40	gridSize = [5 5];	
41		
42	net = selforgmap(gridSize);	
43		
44	%% Step 4: Train SOM	
45	net.trainParam.epochs = 200;	
46		
47	[net,tr] = train(net,X);	
48		
49	%% Step 5: Find Best Matching Unit	
50	outputs = net(X);	
51		
52	[~, neuronIndex] = max(outputs, [], 1);	
53		
54	%% Step 6: Display SOM topology	
55	figure;	
56		
57	plotsomtop(net);	
58		
59	title('SOM Network Topology');	
60		
61	%% Step 7: Display SOM sample hits	
62	figure;	
63		
64	plotsomhits(net,X);	
65		
66	title('SOM Sample Hits');	
67		
68	%% Step 8: Display SOM weight positions	
69	figure;	
70		
71	plotsompos(net,X);	
72		
73	title('SOM Weight Positions');	
74		
75	%% Step 9: Display cluster assignments	
76	disp('Neuron assignment for first 20 samples:');	
77		
78	disp(neuronIndex(1:20));	







Network Diagram

Training Results

Training finished: Reached maximum number of epochs 

Training Progress

Unit	Initial Value	Stopped Value	Target Value	
Epoch	0	200	200	▲
Elapsed Time	-	00:00:09	-	●
				▼

Training Algorithms

Data Division: Batch Weight/Bias Rule trainbu

Performance: Mean Squared Error mse

Calculations: MATLAB

Training Plots

SOM Topology

SOM Neighbor Connections

SOM Neighbor Distances

SOM Input Planes

SOM Sample Hits

SOM Weight Positions